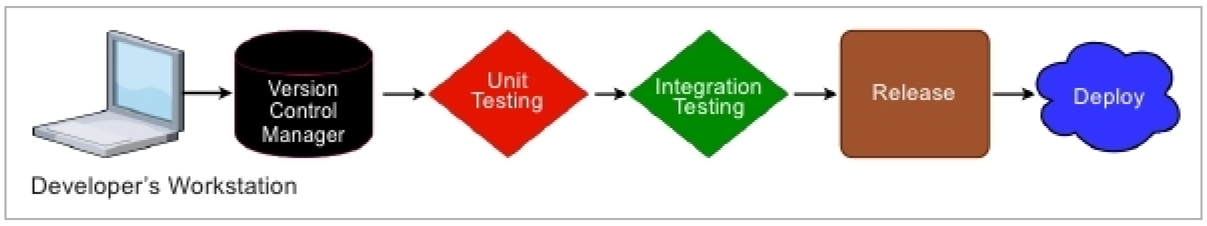

Figure 3: The CI/CD pipeline

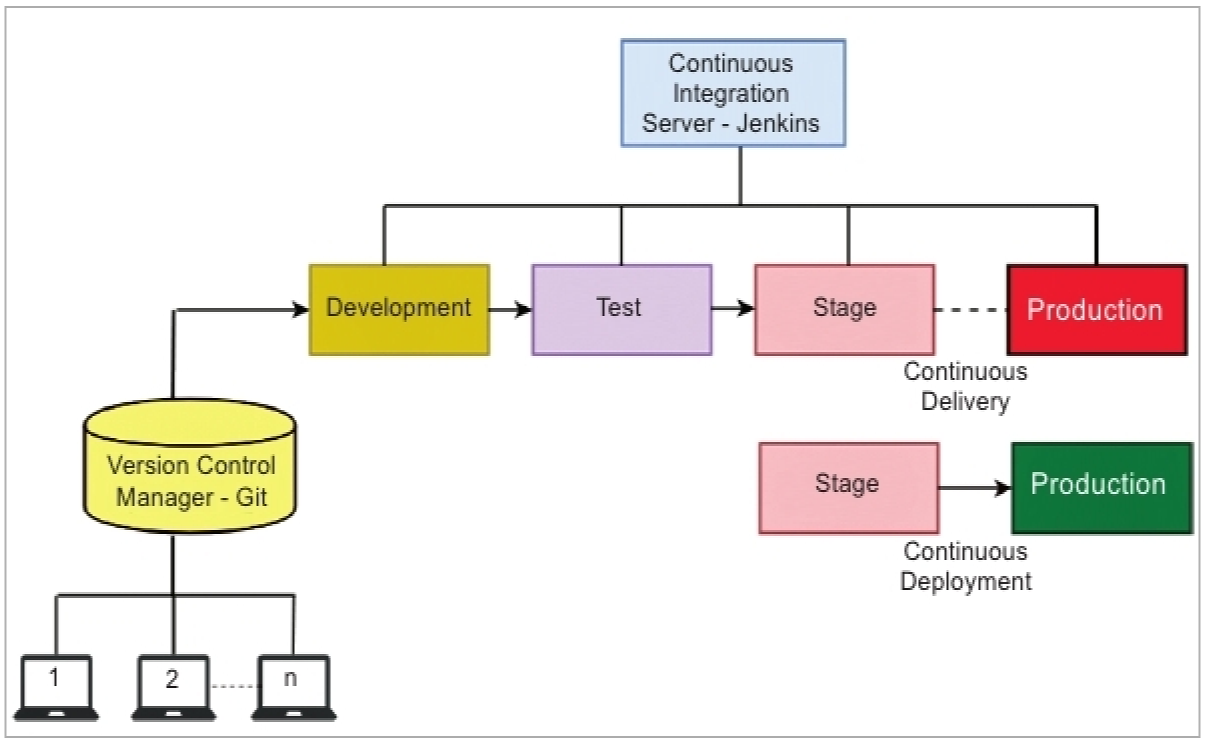

Figure 4: Continuous delivery and continuous deployment

have to adopt CDE and CD. To enable continuous deployment, we have to use automation tools to deploy the changes to production automatically, after every update. CD is employed by many leading companies such as Facebook, Netflix and GitHub.

Some of the many benefits of CD are improved efficiency and productivity of the organisation, faster time to production and reliable/stable releases of the software. CD also makes it easier to scale the systems, because the deployment processes are completely automated. It reduces the risk and the costs involved in moving the changes from the development environment to the production environment. Using CD, companies can even release thousands of stable software updates per day. For example, on an average, Amazon deployed a new software update every 11.7 seconds in 2015, which approximates to 7400 production releases per day. This is possible only by using continuous integration, continuous delivery, continuous deployment and continuous monitoring, which are collectively known as continuous practices.

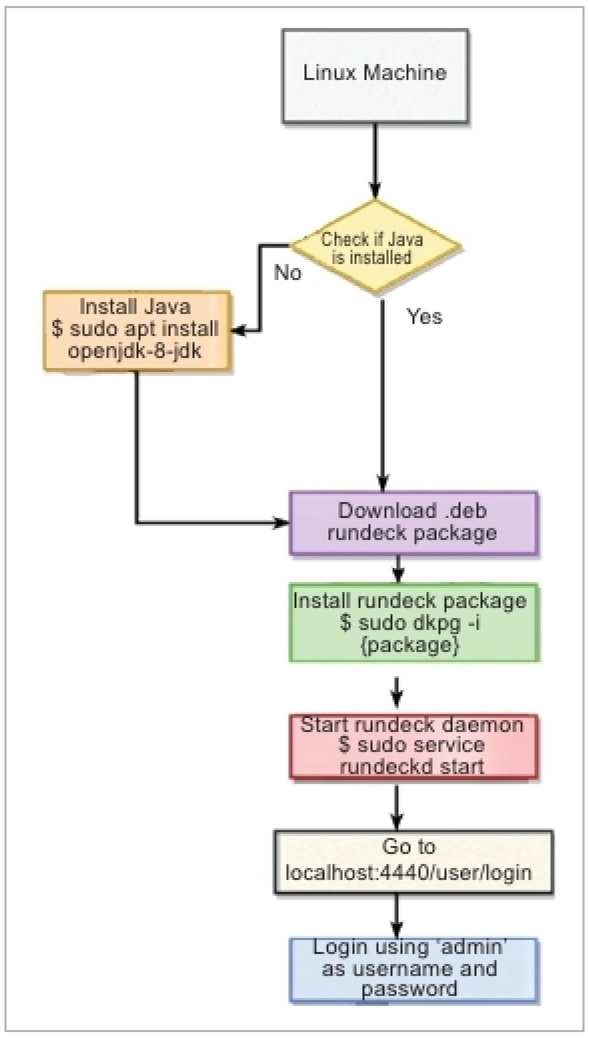

Figure 5: Rundeck installation process

Many IT companies have built their own proprietary deployment tools. Examples are Urbancode Deploy and Urbancode Release by IBM, Codar by HP, Code Stream by VMware, AWS CodeDeploy by Amazon and Visual Studio Release Management by Microsoft. Now, let us look at an open source tool called Rundeck, which enables the automation of processes and tasks on a large scale.

Rundeck

Rundeck is an orchestration tool for continuous deployment. It helps in standardising and automating complex release pipelines and deployment processes. It is an open source Web application written in Java and JavaScript. Rundeck source code is available on GitHub at *https://github.com/rundeck/rundeck*. Many prominent multinational companies such as Walt Disney use Rundeck to carry out their operations. It is an easy-to-use tool that can be accessed via a Web interface, APIs or the command line.

Rundeck is to be installed on a server that is interacting with multiple other hosts. In Rundeck, the nodes are the systems where the operations tasks are to be performed and jobs are the workflows or the processes that need to be carried out on the nodes. The specifics of the job and information about the nodes on which the jobs should be run can be configured through Rundeck. The Web interface of Rundeck also displays the estimated time for the jobs to complete, logs of the jobs and summarised reports of the jobs on each of the nodes. It also has role-based access control policies where only specific users can perform specific jobs.

Rundeck has many plugins to integrate with other continuous practice tools such as Jenkins, Chef, Ansible, Logstash, Puppet, etc. Using these plugins, jobs can be triggered automatically from the continuous integration tools and from the configuration management tools. Also, the jobs can be scheduled based on the application's requirements. Rundeck can send notifications to the developers